

Ext

8

Transformers and Transmission of Electricity

In this chapter, we will learn about

- how a simple transformer works
- how to improve the efficiency of a transformer
- how electrical energy is transmitted over long distances

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8.1 Transformers

When you connect a laptop computer to a mains socket, the connection is via a box-like object (Fig 8.1a). What is it?



Fig 8.1a *A laptop computer is connected to the mains socket via a box-like object.*

It is a **transformer**. It converts the mains voltage (220 V) to the operating voltage (19 V) of the computer. Many appliances have their own transformers so that they can work at the desired voltages when connected to the mains sockets. Let us learn more about transformers.

Flipped classroom
Watch a video and answer the questions.

→ This simulation shows (a) how switching on and off a d.c. in the primary coil produces an induced current in the secondary coil, (b) how an a.c. in the primary coil produces an induced a.c. in the secondary coil.

DSE goal
Describe the structure of a simple transformer and how it works.

Simulation Video

→ This video demonstrates the working principle of a transformer.

1 Working principle

A simple transformer consists of two separate coils wound around a soft-iron core (Fig 8.1b). Coil *X* is connected to an a.c. power supply and is called the **primary coil**. Coil *Y* is connected to a load and is called the **secondary coil**.

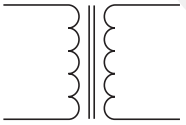
A transformer also works with a varying d.c. ▶

When an a.c. passes through *X*, a varying magnetic field is produced. When *Y* experiences this varying magnetic field, an alternating e.m.f. is induced across *Y*. This e.m.f. drives an a.c. through the load. Therefore, electrical energy is transferred from the primary circuit to the secondary circuit.

The a.c. in the two coils have the same frequency. ▶

Figure 8.1c shows the circuit symbol of a transformer.

The circuit symbol can also be drawn with two lines in the middle. ▶



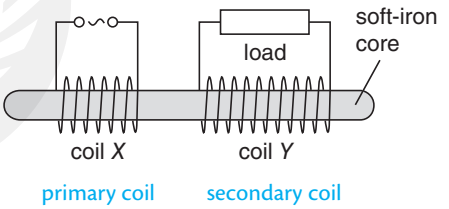


Fig 8.1b *A simple transformer.*

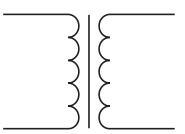


Fig 8.1c *Circuit symbol of a transformer.*

The size of the induced e.m.f. across *Y* depends on the numbers of turns of the two coils. Let us investigate this in Experiment 8a.

transformer 變壓器（俗稱火牛）　primary coil 原線圈　secondary coil 副線圈

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DSE goal

Relate the voltage ratio to turns ratio by $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ and apply it to solve problems.

DSE exam

12(1B)Q9, 18(1A)Q30, 19(1A)Q29, 21(1A)Q30 23(1A)Q29

2 Voltage ratio and turns ratio

Experiment 8a / Voltage ratio and turns ratio

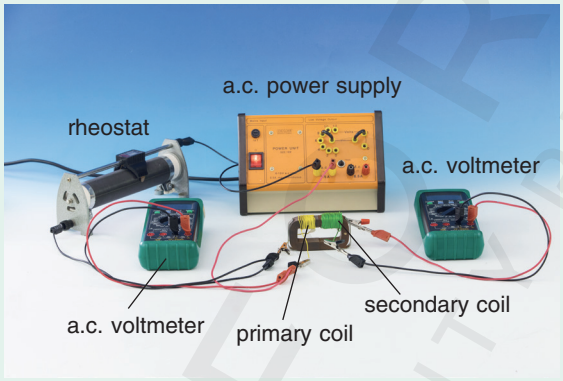
1 Build a transformer by winding 20 turns in the primary coil and 60 turns in the secondary coil around a soft-iron core.

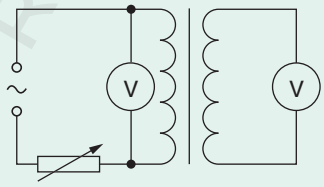
2 Connect the primary coil to an a.c. power supply via a rheostat.

3 Connect an a.c. voltmeter across each coil.

An a.c. voltmeter gives the r.m.s. value of the voltage.

The experiment can also be conducted with voltage sensors instead of a.c. voltmeters.





Video

→ This video demonstrates Expt 8a.

4 Turn on the power supply. Record the voltage V_p across the primary coil and the voltage V_s across the secondary coil.

5 Repeat the experiment using secondary coils with different numbers of turns.

Ss may obtain a voltage ratio slightly larger than the turns ratio. Ts may explain that not all the magnetic field lines pass through the secondary coil.

Further investigation can be done by replacing the soft-iron C-cores with wooden C-cores.

Discussion

What is the relationship between the ratio of the voltages across the two coils and the ratio of the numbers of turns in the two coils?

The voltage across the primary coil is the primary voltage, denoted by V_p, and that across the secondary coil is the secondary voltage, denoted by V_s. The number of turns in the primary coil is denoted by N_p and that in the secondary coil is denoted by N_s (Fig 8.1d).

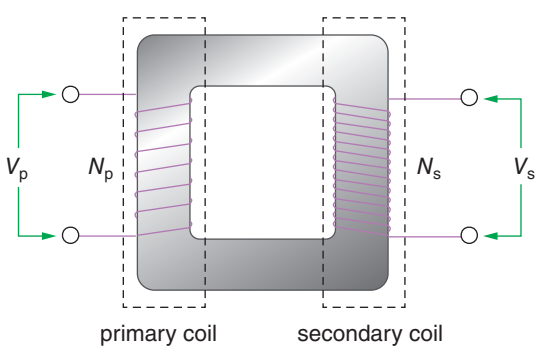


Fig 8.1d *Voltages and numbers of turns of a transformer.*

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Simulation

→ This simulation shows how the turns ratio of a simple transformer is related to the voltage ratio.

In the experiment, the **voltage ratio** $V_p : V_s$ is about the same as the **turns ratio** $N_p : N_s$. In general, if all the magnetic field lines go through both coils, i.e. **no flux leakage**,

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

V_p: primary voltage (V); V_s: secondary voltage (V); N_p: number of turns in primary coil (no unit); N_s: number of turns in secondary coil (no unit)

The formula holds for both peak values and r.m.s. values.

Soft-iron cores are usually used to prevent flux leakage. We shall assume that this formula can be applied **in all cases** in this book.

By setting the turns ratio, we can change an a.c. voltage to a higher or a lower value, i.e. to step up or step down the voltage (Table 8.1a).

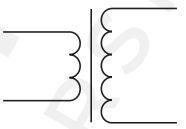
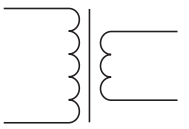
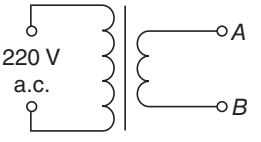
	Step-up transformer	Step-down transformer
Voltage	$V_p < V_s$	$V_p > V_s$
Number of turns	$N_p < N_s$	$N_p > N_s$
Symbol		

Table 8.1a *Step-up transformer and step-down transformer.*

Example 1 Turns ratio and voltage ratio

The primary coil of a transformer is connected to the 220-V mains. There are 2200 turns in the primary coil and 110 turns in the secondary coil.



(a) What is the voltage across the secondary coil AB?

(b) If the primary coil is connected to a steady d.c. supply of 20 V, what will be the voltage across AB?

Solution

(a) By $\frac{V_p}{V_s} = \frac{N_p}{N_s}$,

$$V_s = \frac{N_s}{N_p} \times V_p = \frac{110}{2200} \times 220 = 11 \text{ V}$$

(b) 0

Cross-link
We learn in Ch 4.3 that 220 V is the r.m.s. value of the mains voltage.

Tip
The magnetic field produced by the primary coil will be steady, so no e.m.f. is induced in the secondary coil.

voltage ratio 電壓比　turns ratio 匝數比　step-up transformer 升壓器
step-down transformer 降壓器

Ext

Google Forms

Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) In a transformer, the secondary voltage is always lower than the primary voltage. (T/F)

(b) A transformer can work with d.c. (T/F)

A transformer can work with a varying d.c.

2 Which of the following can increase the output voltage of a transformer? Put a tick in the corresponding box(es).

☐ Increase the voltage of the input a.c.

☐ Increase the number of turns in the primary coil.

☐ Increase the number of turns in the secondary coil.

3 Complete the following table:

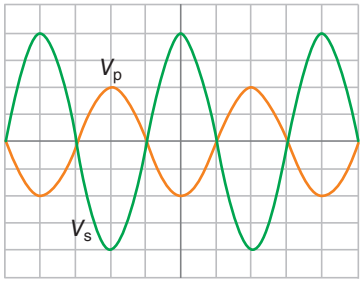
	V_p / V	V_s / V	N_p	N_s	Type of transformer
(a)	5	500	200		step-up
(b)	220		1000	100	

Learn by practice 1

1 The primary coil of a transformer is connected to a 20-V a.c. supply. The voltage across the secondary coil is 100 V. Find the turns ratio of the transformer.

2 A lamp is connected to an a.c. power supply via a transformer. It operates at its rated values '8 V, 12 W'. The primary coil of the transformer has 2750 turns. The secondary coil has 100 turns. Find the output voltage of the power supply.

3 The primary coil of a transformer is connected to an a.c. supply. The primary voltage V_p and the secondary voltage V_s are as shown. What is the turns ratio of the transformer?



V_s depends on V_p and the turns ratio, while V_p is determined solely by the voltage supplied.

▶ To provide different output voltages, some transformers are multi-tapped or with multiple winding (Fig 8.1e).

The low voltage power pack commonly found in a school laboratory contains multi-tapped transformer to give different voltages.

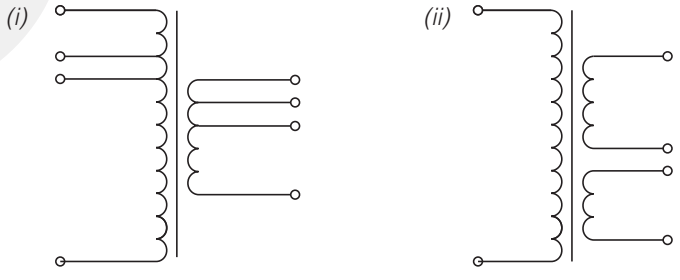
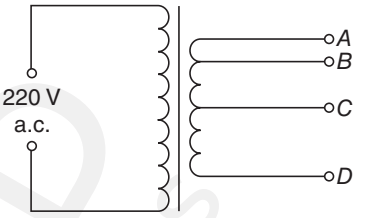


Fig 8.1e *(i) A multi-tapped transformer and (ii) a multiple-winding transformer.*

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Example 2 | A multi-tapped transformer

The primary coil of a transformer is connected to the 220-V mains. There are 2200 turns in the primary coil. The secondary coil is tapped such that the number of turns across *AB*, *BC* and *CD* are 30 turns, 40 turns and 50 turns respectively.



(a) What is the voltage across *AB*?

(b) A device works at 7 V. Which terminals should this device be connected to?

Solution

(a) By $\frac{V_p}{V_s} = \frac{N_p}{N_s}$,

$$V_s = \frac{N_s}{N_p} \times V_p = \frac{30}{2200} \times 220 = 3 \text{ V}$$

Tip
Consider the part across *AB* in the secondary coil.

(b) $$N_s = \frac{V_s}{V_p} \times N_p = \frac{7}{220} \times 2200 = 70$$

Tip
Check out across which two taps the number of turns is 70.

$\therefore$ The device should be connected to terminals *A* and *C*.

Know more

Shaver socket

Besides changing a voltage from one value to another, a transformer can also be used as a safety device. It is commonly used in the shaver socket in a hotel bathroom (Fig a).

A bathroom may be full of water vapour, which may connect a user's hand to the wire in the socket and cause an electric shock. To prevent such an accident, a shaver socket has a transformer whose secondary coil is not earthed (Fig b). The socket holes are connected to the secondary coil. If a user is connected to a point on the secondary coil by water vapour, no current will pass through him because the circuit is not completed.

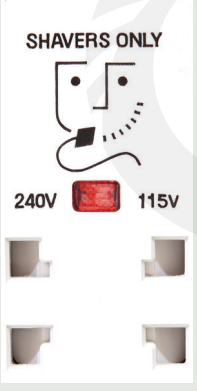


Fig a

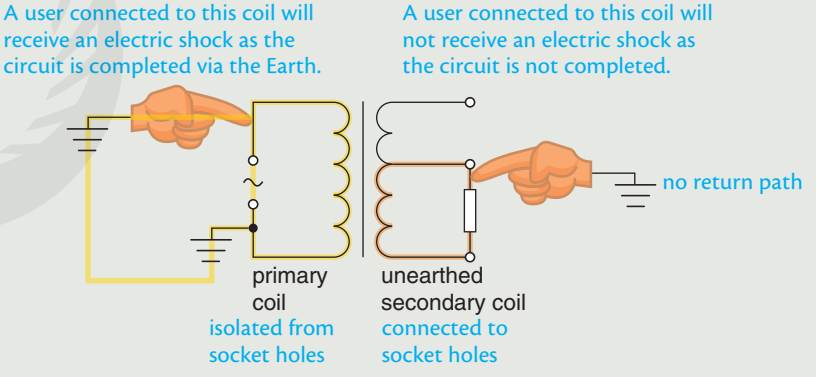


Fig b

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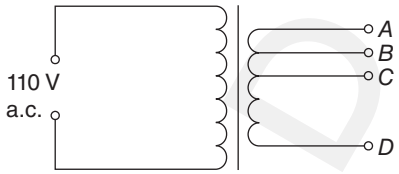
Google Forms

Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 The secondary coil of a transformer is tapped at different points as shown. The primary coil has 5500 turns and is connected to a 110-V a.c. power supply.



Complete the table below.

	Secondary coil		
	Part	**Number of turns**	**Voltage**
(a)	across *AD*	1100	
(b)	across *AB*		5 V
(c)	across *BC*	440	
(d)	across *CD*		

DSE goal

Examine methods for improving the efficiency of a transformer.

DSE exam

15(1B)Q8(b)(iii)

3 Practical transformers

Figure 8.1f shows two possible arrangements of coils in practical transformers. Both designs ensure that the magnetic field is trapped in closed loops in the core to avoid flux leakage.

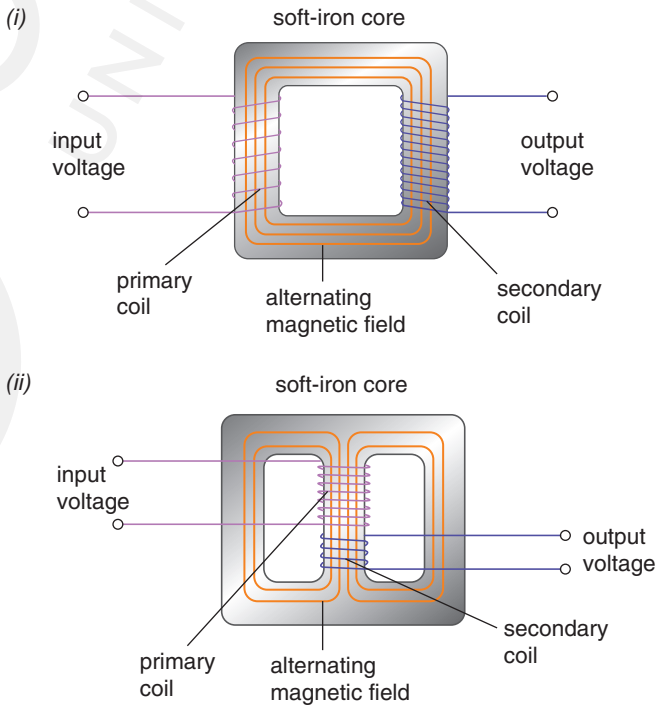


Fig 8.1f *Two kinds of winding of practical transformers.*

Ext

See the next page for the definition of efficiency. ▶

Power loss is unavoidable in practical transformers. Table 8.1b shows the sources of power loss and the measures to reduce the loss. A well-designed transformer can have an **efficiency** over 90%. Note that the equation $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ is still valid for practical transformers.

May ask Ss what else can be done to reduce the resistance of wires. Some possibilities are:
1 keeping the wire cool;
2 keeping the length of wire short;
3 using a more conductive material for the wire.
Further discussion may be made on the feasibility of the possibilities.

Source	Measure
Resistance of wires: The wires winding the coils have resistance and become heated when currents flow through them. The power dissipated this way is called **ohmic loss**.	Use **thick wires** to keep the coil resistance low.
Magnetization and demagnetization of the core: Energy is lost as heat when the core is magnetized and demagnetized repeatedly.	Use cores made of **soft iron** which can be magnetized and demagnetized easily.
Eddy currents in the core: The changing magnetic field produced by the primary current induces eddy currents in the core. Energy is dissipated as heat. This power loss is called **eddy current loss**.	Use a **laminated** core which consists of thin sheets of soft iron, each with insulting coating (Fig 8.1g). This reduces eddy currents as they are limited to flow within each sheet (Fig 8.1h).

Table 8.1b *Sources of power loss in a transformer and the measures to reduce the loss.*

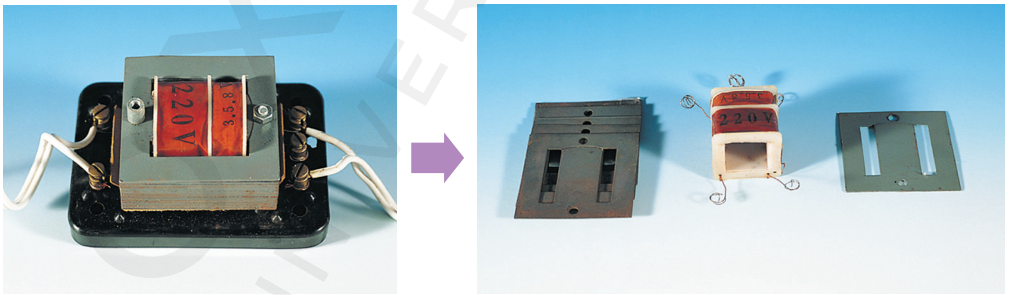


Fig 8.1g *A practical transformer with a laminated core.*

Know more

Electronic transformers

The plug of a smartphone contains an electronic transformer, which is much lighter and smaller than a conventional one.



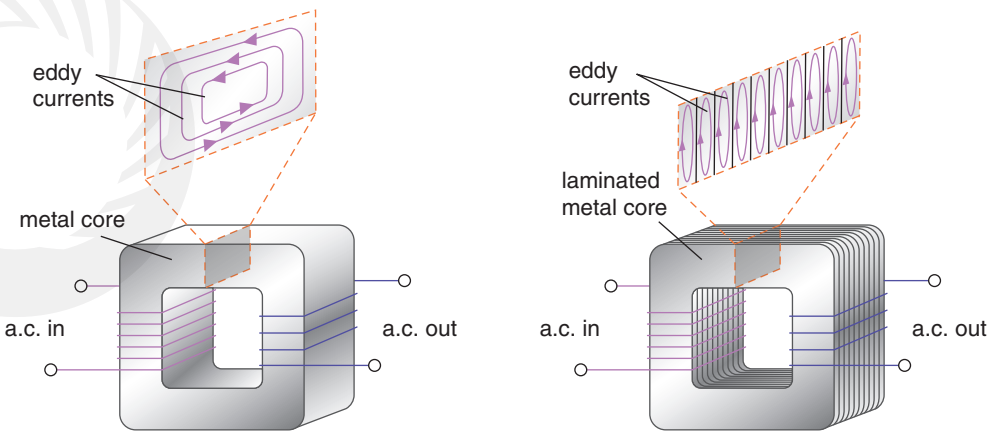


Fig 8.1h *Eddy currents are reduced by using a laminated core. The planes of the laminations are arranged parallel to the B- field.*

efficiency 效率　　laminated 疊片的

Text

DSE exam

16(1A)Q31, 17(1A)Q30, 20(1A)Q29

4 Efficiency of practical transformer

If a transformer is **ideal**, all the power supplied to the primary circuit can be transferred to the secondary circuit. That is,

power input to primary coil P_p = power output from secondary coil P_s

Recall that $P = VI$.

primary voltage (V) — primary current (A) — secondary voltage (V) — secondary current (A)

$$V_p I_p = V_s I_s$$

From this equation, we have

$$\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

However, there is always power loss in **practical** transformers. The efficiency of a practical transformer is given by

$$\text{efficiency} = \frac{P_s}{P_p} \times 100\% = \frac{V_s I_s}{V_p I_p} \times 100\%$$

$$\overbrace{V_p I_p}^{P_p} \times \text{efficiency} = \overbrace{V_s I_s}^{P_s}$$

Caution

When the transformer is not ideal,

$\frac{V_p}{V_s} = \frac{N_p}{N_s}$

$\frac{I_s}{I_p} \neq \frac{V_p}{V_s}$

Example 3 Efficiency of a transformer

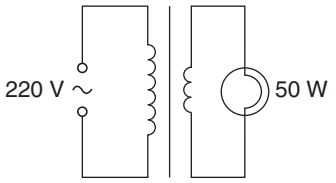
A light bulb is connected to the 220-V mains via a transformer. The turns ratio of the transformer is 11 : 1. The power consumed by the bulb is 50 W. The primary current is 0.3 A.

(a) Find the secondary voltage.

(b) Find the secondary current.

(c) Find the power output of the mains.

(d) Find the efficiency of the transformer.



This is equal to the power input to the primary coil.

Solution

(a) $V_s = \frac{N_s}{N_p} \times V_p = \frac{1}{11} \times 220 = 20$ V

As the voltage is stepped down, the current is stepped up.

(b) $I_s = \frac{P_s}{V_s} = \frac{50}{20} = 2.5$ A

(c) $P_p = V_p I_p = (220)(0.3) = 66$ W

(d) Efficiency $= \frac{P_s}{P_p} \times 100\% = \frac{50}{66} \times 100\% = 75.8\%$

Ext

Google Forms

Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Complete the following table about the efficiency of transformers.

	Efficiency	V_p / V	V_s / V	I_p / A	I_s / A
(a)	100%		500	5	1
(b)	90%	10	200		2
(c)		220	20	0.1	0.99

2 The power loss of a transformer can be reduced by using

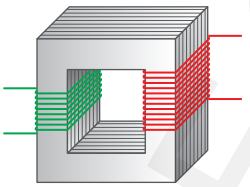
(a) (thicker / thinner) wires to make the coil,

(b) (steel / soft iron) to make the core,

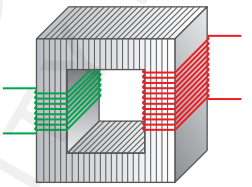
(c) a (solid / laminated) core.

3 How should the core of a transformer be laminated for a high efficiency?

A



B



4 Which of the following must be correct for transformers with no flux leakage?

A $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

B $\frac{I_p}{I_s} = \frac{N_s}{N_p}$

C $\frac{I_p}{I_s} = \frac{V_p}{V_s}$

Learn by practice 2

1 A video game console consumes a power of 15 W when it is connected to the 220-V mains via an ideal transformer. Find the current drawn from the mains.

2 A load is connected to a transformer. The primary voltage is 240 V. The primary current is 2 A. The secondary voltage is 100 V. The secondary current is 4 A.

(a) Find the efficiency of the transformer.

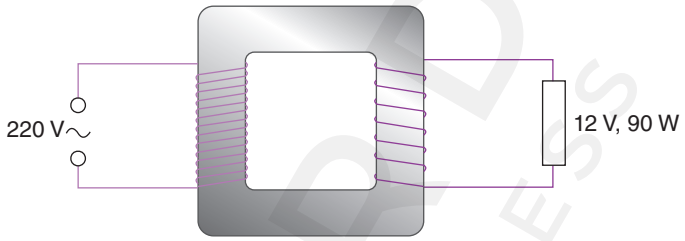
(b) Suggest two designs of a transformer that can improve its efficiency.

3 A load is connected to an a.c. power supply via a transformer. The primary voltage is 110 V. The primary current is 2 A. The turns ratio is 1 : 2. The efficiency of the transformer is 80%. Find the secondary current.

Ext

Example 4 | Ideal and practical transformers

A load rated at '12 V, 90 W' is connected to the 220-V mains via a transformer. The primary coil has 4000 turns. The secondary coil has 200 turns.



(a) Suppose the transformer is ideal.

(i) Find the secondary current.

(ii) Find the primary current.

(b) Suppose the efficiency of the transformer is 50%.

(i) Find the secondary current.

(ii) Find the primary current.

(c) Suggest and explain one method to improve the efficiency of a practical transformer.

Solution

Tip
The load does not work at its rated voltage.

Tip
R can be found using the rated values.

Tip
Since the turns ratio and primary voltage remain unchanged, the secondary voltage, hence the secondary current, also remain unchanged.

A larger current, hence a higher power, in the primary circuit is needed to operate the load when the transformer is not ideal.

(a) (i) $V_s = \frac{N_s}{N_p} \times V_p = \frac{200}{4000} \times 220 = 11$ V

Resistance R of load $= \frac{V^2}{P} = \frac{12^2}{90} = 1.6\ \Omega$

$I_s = \frac{V_s}{R} = \frac{11}{1.6} = 6.88$ A

(ii) $I_p = \frac{N_s}{N_p} \times I_s = \frac{200}{4000} \times 6.88 = 0.344$ A

(b) (i) 6.88 A

(ii) $V_p I_p \times$ efficiency $= V_s I_s$

$220 I_p \times 50\% = (11)(6.88)$

$I_p = 0.688$ A

(c) Any one of the following:

Use a soft iron core to reduce the energy loss in magnetizing and demagnetizing the core.

Use a laminated core to reduce eddy currents in the core.

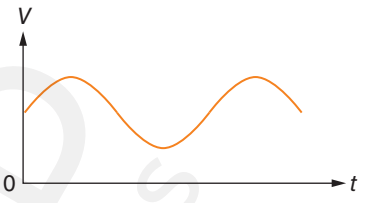
Use thicker wires for the coils to reduce the resistance of the coils.

Exercise 8.1 Q4 (p.334)

Ext

Writing practice

The primary coil X of a transformer is connected to a power supply and the secondary coil Y is connected to a load. The output voltage V of the power supply varies with time t as shown.



Question 1

Does a current pass through the load? Explain. (3 marks)

Answer

The varying current through X generates a varying magnetic field. 1A

As Y experiences this varying magnetic field, 1A

an e.m.f. is induced in Y and drives a current through the load. 1A

Question 2

If another load is connected in parallel to the original one and the efficiency of the transformer remains unchanged, how will the primary current change? Explain. (2 marks)

Answer

The original load will consume the same power and the new load will also consume a certain power. Therefore, the power consumed P_s in the secondary coil will increase. 1A

Or

The total resistance in the secondary circuit will decrease. By $P = \frac{V^2}{R}$, the power consumed P_s in the secondary coil will increase. 1A

The current and power output from the power supply depend on the power consumed by the load. ▶

By $V_p I_p \times$ efficiency $= P_s$, since the efficiency and V_p do not change, the primary current I_p will increase when P_s increases. 1A

Practise yourself

1 The primary coil of a transformer is connected to a battery via a switch. Will there be a voltage across the secondary coil when the switch is just closed? Explain. (3 marks)

2 The primary coil of a transformer is connected to the mains. The secondary coil is connected to a load. If one more load is connected in series to the original one, how will the primary current change? Assume the efficiency of the transformer remains unchanged. (2 marks)

Ext

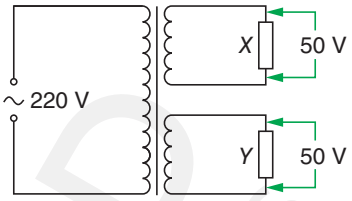
☆ HKDSE 2017 Paper 1A Q30, 2020 Paper 1A Q29

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | A multiple-winding transformer

A multiple-winding transformer is connected to two loads X and Y as shown. The efficiency of the transformer is 90%. When the primary coil is connected to the 220-V mains, the primary current is 2 A. The voltage across X is 50 V and the current through it is 4.8 A. The voltage across Y is also 50 V. What is the current through Y?



A 2.8 A B 3.12 A

C 4.8 A D 4.97 A

Solution

$P_p \times \text{efficiency} = P_s$

$V_p I_p \times \text{efficiency} = V_X I_X + V_Y I_Y$

$(220)(2) \times 90\% = (50)(4.8) + 50 I_Y$

$I_Y = 3.12 \text{ A}$

∴ The answer is B

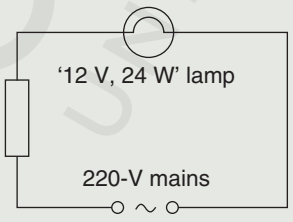
Common mistake ⚠
Students may not be able to find the output power when there is more than one load.

▶ Revision 8 Q11 (p.343)

Know more

Why is a transformer preferred?

A '12 V, 24 W' lamp can be operated by connecting it in series with a suitable resistor to the 220-V a.c. mains. However, the efficiency of this method is very low and therefore a transformer is preferred.



Operating current $= \frac{24}{12} = 2$ A

Input power $= 220 \times 2 = 440$ W

Efficiency $= \frac{24}{440} \times 100\% = 5.45\%$

Try Revision 8 Q18 (p.345). ▶

Exercise 8.1

★1

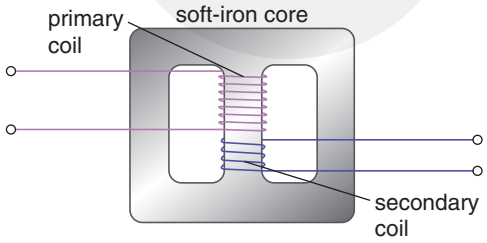


Fig a

Which of the following measures can improve the efficiency of the transformer in Figure a?

(1) Increase the turns ratio.

(2) Use a laminated soft-iron core.

(3) Wind the secondary coil on the right arm of the soft-iron core.

A (2) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

Ext

*2 Which of the following statements about a transformer is/are correct?

(1) No magnetic field is produced if a steady d.c. passes through the primary coil.

(2) An e.m.f. is induced in the secondary coil as long as the secondary coil experiences a change in magnetic flux.

(3) The transformer only works with a.c.

A (1) only

B (2) only

C (1) and (2) only

D (1) and (3) only

3 Figure b shows a transformer used for charging a mobile phone. It can operate with an input voltage of 100–240 V and its output voltage and current are always 4.8 V and 0.9 A respectively. Suppose its efficiency is 90%. Find the primary current when it is connected to the 100-V mains.

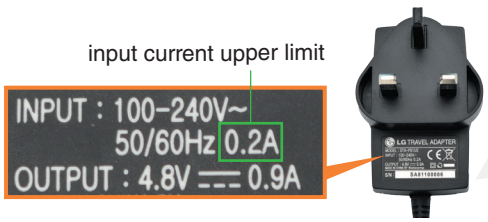


Fig b

*4 When a light bulb is connected to the 220-V mains through an ideal transformer, the p.d. across the bulb is 10 V and a current of 0.5 A passes through it (Fig c).

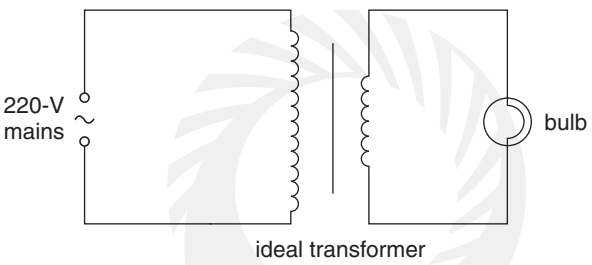


Fig c

(a) What is the turns ratio of the transformer?

(b) What power is consumed by the bulb?

(c) What is the power output from the mains?

(d) What current is drown from the mains?

(e) If the transformer is 85% efficient, what will be the current drawn from the mains?

*5 A light bulb (22 V, 25 W) is powered by the 220-V mains via a transformer of efficiency 90%.

(a) If the light bulb works at its rated voltage, find the turns ratio $N_p : N_s$.

(b) Find the current drawn from the mains.

(c) If another light bulb (22 V, 15 W) is connected (i) in parallel, (ii) in series, to the first one, find the new current drawn from the mains.

(d) Suggest a method that can improve the efficiency of a transformer.

*6 The primary and secondary voltages of a transformer are displayed by CROs as shown (Fig d). The Y-gain (scale along the vertical direction) is set at 5 V / div in each graph. Assume that the transformer is ideal.

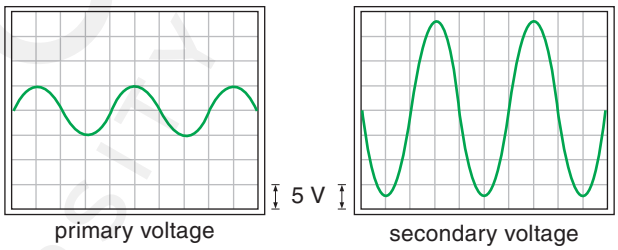


Fig d

(a) Explain how an e.m.f. is induced in the secondary coil.

(b) If the primary coil has 100 turns, what is the number of turns in the secondary coil?

(c) Suggest how the transformer should be modified so that the induced e.m.f. in the secondary coil is increased.

*7 Figure e shows a transformer. Coil *AB* has 600 turns. Coil *CD* has 200 turns. Coil *EF* has 40 turns. Assume the transformer is ideal.

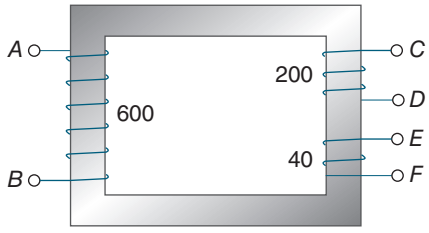


Fig e

(a) If a 24-V a.c. power supply is connected across *AB*, what is the voltage across *CD* and *EF* respectively?

(b) If a 24-V a.c. power supply is connected across *EF*, what is the voltage across *AB* and *CD* respectively?

Ext

8.2 Power transmission

You may have seen on the street facilities with high voltage warning signs.



This facility is a substation. See p.338.

Fig 8.2a *A facility dealing with a high voltage.*

Do you know why electricity is transmitted at a high voltage?

DSE goal

Discuss the advantages of transmission of electrical energy with a.c. at high voltages.

DSE exam

15(1B)Q8

1 Power loss in transmission cables

Electricity is generated in power stations, which are in remote areas. It has to be transmitted over great distances from the power stations to users via transmission cables (also called power lines) (Fig 8.2b). The long cables have resistance and the power loss in them can be significant.

When current I passes through cables of resistance R, power loss = I^2R

To reduce this power loss, electricity is transmitted at a high voltage. Let us demonstrate this in Experiment 8b.



Fig 8.2b *Electrical energy is transmitted via transmission cables.*

→ This video demonstrates Expt 8b.

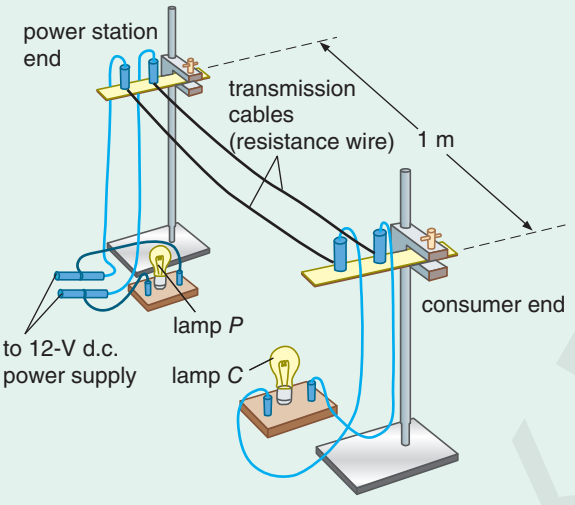
Ext

Experiment 8b / Transmission of electricity

Video

1 Set up the model transmission cables with two '12 V, 24 W' lamps (Fig a). Switch on the 12-V d.c. power supply and compare the brightness of the two lamps. The brightness of the lamp *P* indicates the power supplied by the 'station' and the brightness of lamp *C* indicates the power delivered to the 'consumer'. Repeat with a 12 V a.c. power supply.

2 Step up the 12 V a.c. at the 'station' end by a 1 : 20 transformer. Then step it down at the 'consumer' end by a 20 : 1 transformer (Fig b). Switch on the power supply and compare the brightness of the lamps.



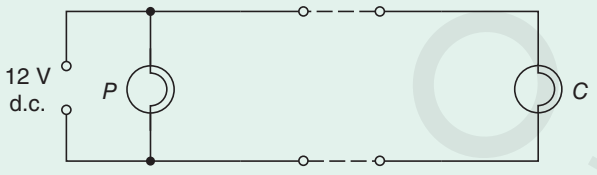
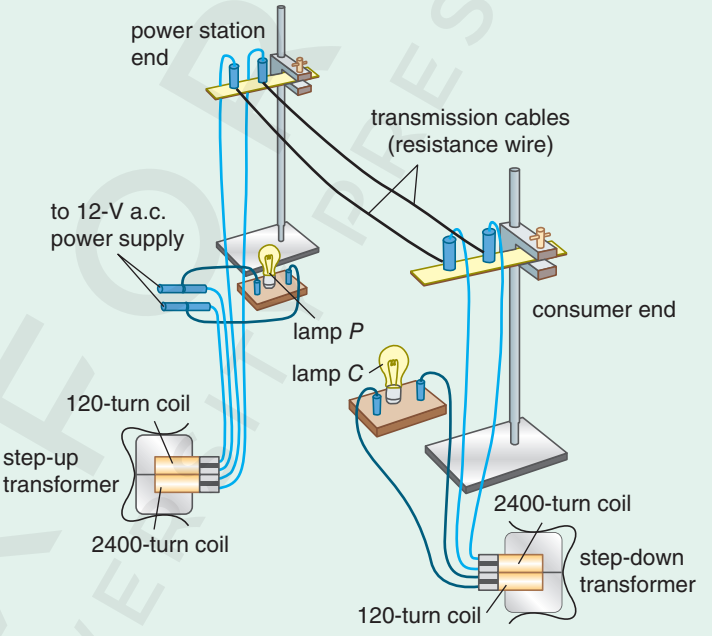


Fig a



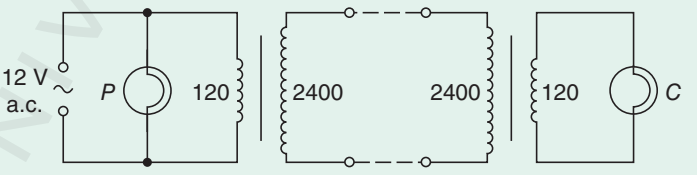


Fig b

Do not touch the transmission cables. They are now at 240 V a.c.!

Discussion

Which setting results in the least power loss in the transmission cables?

In Experiment 8b, when electricity is transmitted at 12 V, lamp *C* is much dimmer than lamp *P*. This shows that the power loss in the cables is large.

▶ When electricity is transmitted at 240 V, lamp *C* is almost as bright as lamp *P*. This shows that the power loss in the cables is low.

$$V_s = \frac{N_s}{N_p} \times V_p$$
$$= \frac{2400}{120} \times 12$$
$$= 240 \text{ V}$$

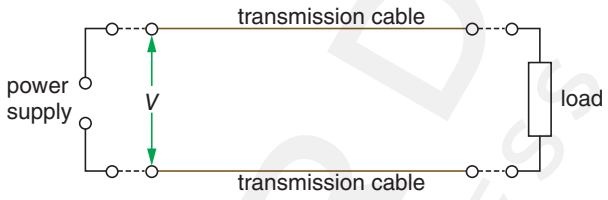
Note that **a.c.** is usually used in transmitting electricity because it **can be stepped up or down easily** by transformers.

The power loss in transmission cables can be reduced if electricity is transmitted at a high voltage.

Ext

Example 5 Power loss in transmission cables

Electrical power of 4 kW is transmitted from a power supply to a load at voltage V via transmission cables with a total resistance of 5 Ω. Find the percentage of power loss during transmission if V is (a) 100 V, (b) 2 kV.



Solution

(a) Current through cables $= \frac{P}{V} = \frac{4000}{400} = 10$ A

Power loss in cables $= I^2R = (10)^2(5) = 500$ W

Percentage of power loss $= \frac{500}{4000} \times 100\% = 12.5\%$

For the same P, $V \uparrow \Rightarrow I \downarrow$ ▶

(b) Current through cables $= \frac{P}{V} = \frac{4000}{2000} = 2$ A

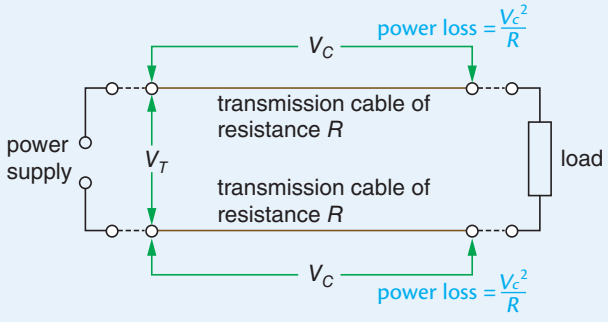
Power loss in cables $= I^2R = (2)^2(5) = 20$ W

Percentage of power loss $= \frac{20}{4000} \times 100\% = 0.5\%$

Right and wrong

✗ By $P = \frac{V^2}{R}$, the power loss in transmission cables is high if electricity is transmitted at a high voltage.

✓ When using $P = \frac{V^2}{R}$ to calculate the power loss in a transmission cable, the term V refers to the voltage across a cable (V_C in the following diagram), not the transmitting voltage V_T. The total power loss in the cables is equal to $\frac{2V_C^2}{R}$ but not $\frac{2V_T^2}{R}$. Increasing V_T can reduce V_C, hence reducing the power loss.



Know more

War of currents

Engineers, including *Thomas Edison* and Nikola Tesla disagreed on whether d.c. or a.c. was better. They competed fiercely.

▶ p.348

Thomas Edison 愛迪生

Ext

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Why is high voltage used in power transmission?

A This reduces the resistance of the transmission cables.

B This reduces the current through the transmission cables.

2 Why is a.c. used in power transmission?

A A.c. can be stepped up or down easily.

B No power is lost when a.c. flows through the transmission cables.

Learn by practice 3

1 Electrical power of 20 MW is transmitted via transmission cables with a total resistance of 2 Ω.

(a) Suppose the power is transmitted at 20 kV.

(i) Find the current through the cables.

(ii) Find the power loss in the cables.

(b) Suppose the power is transmitted at voltage V. The power loss in the cables is 10 kW.

(i) Find the current through the cables.

(ii) Find V.

DSE goal

Describe various stages of stepping up and down of the voltage in a grid system for power transmission.

2 Grid system for power transmission

To distribute electricity to different places, a transmission network called the **grid system** is required (Fig 8.2c). The voltage is first stepped up at the power station before the electricity is transmitted. It is then stepped down several times before the electricity is distributed to factories, offices and homes.

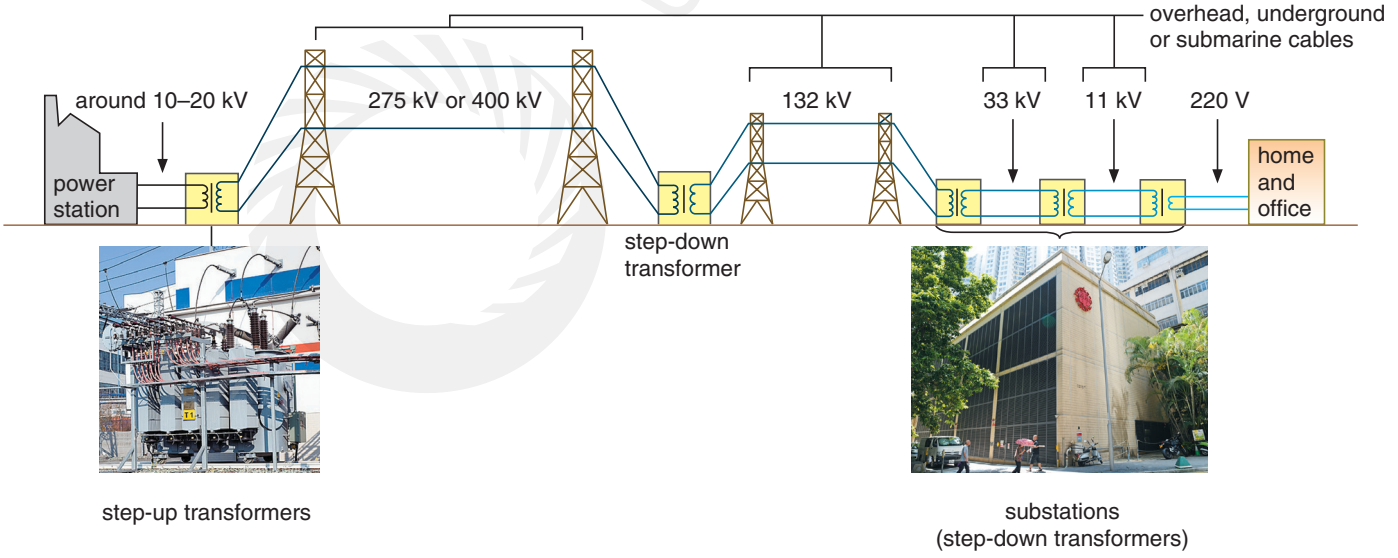


Fig 8.2c *A schematic diagram showing the transmission of electricity from power station to home and office.*

grid system 電網系統

Ext

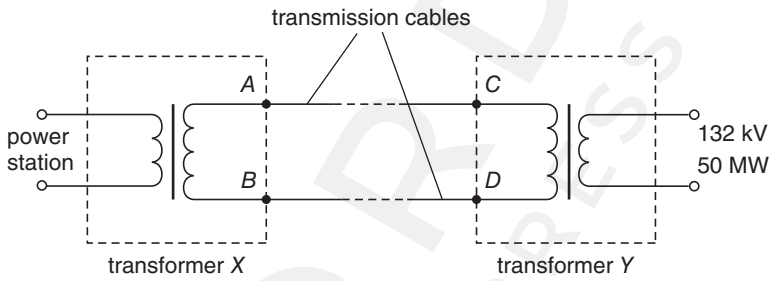
STSE

Electricity supply in Hong Kong

Electricity supply in Hong Kong is provided by two companies. *HK Electric* serves the Hong Kong Island, Lamma Island and Ap Lei Chau. *CLP* serves Kowloon, New Territories, Lantau and other outlying islands.

Example 6 A power transmission system

The figure shows a simplified power transmission system. Each transmission cable is 20 km long and its resistance is 0.1 Ω per km. Transformers *X* and *Y* are ideal. The voltage and power output of transformer *Y* are 132 kV and 50 MW respectively. The turns ratio of transformer *Y* is 3 : 1.



(a) Find the p.d. across *C* and *D*.

(b) Find the total power loss in the transmission cables.

(c) Find the efficiency of this transmission system.

(d) Find the p.d. across *A* and *B*.

(e) Suggest two ways to increase the efficiency of this transmission system.

Solution

(a) Consider transformer *Y*.

$$V_{CD} = V_p = \frac{N_p}{N_s} \times V_s = \frac{3}{1} \times 132 \times 10^3 = 396 \times 10^3 \text{ V} = 396 \text{ kV}$$

(b) Since transformer *Y* is ideal,

$P_p = P_s = 50$ MW

$$\text{Current through cables} = I_p = \frac{P_p}{V_p} = \frac{50 \times 10^6}{396 \times 10^3} = 126.3 \text{ A}$$

Power loss in each cable $= I_p^2 R = (126.3)^2(0.1 \times 20) = 31\,900$ W

Total power loss in cables $= 2 \times 31\,900 = 63\,800$ W

(c) Power input to system $= 50 \times 10^6 + 63\,800 = 5.01 \times 10^7$ W

$$\text{Efficiency of transmission system} = \frac{50 \times 10^6}{5.01 \times 10^7} \times 100\% = 99.8\%$$

(d) Voltage drop across each cable $= I_p R = (126.3)(0.1 \times 20) = 253$ V

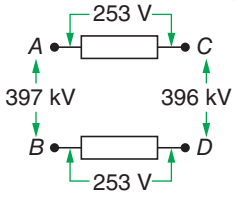
$V_{AB} = 396 \times 10^3 + 253 + 253 = 397 \times 10^3$ V $= 397$ kV

(e) Transmit the power through the cables at a higher voltage.

Use cables of lower resistance.

Exercise 8.2 Q4 (p.340)

Tip



Ext

Exercise 8.2

1 A simplified power transmission system is shown (Fig a). Suppose transformers T_1 and T_2 are ideal.

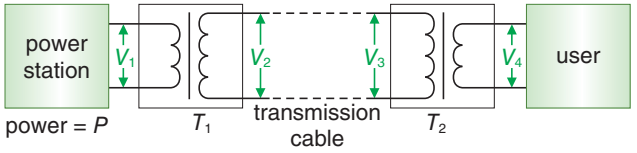


Fig a

Which of the following regarding the system must be correct?

A $V_1 = V_4$

B $V_3 > V_2$

C The system is 100% efficient.

D None of the above

2 Is a.c. or d.c. usually used in transmitting electricity?

A A.c. is used because a power station can only generate a.c.

B A.c. is used because it can be stepped up or down easily.

C D.c. is used because it is safer.

D D.c. is used because it flows faster through the cables.

***3** Which of the following measures can reduce the power loss in transmission cables?

(1) Increase the current through the cables.

(2) Transmit electricity at a higher voltage.

(3) Use thicker cables which are made of the same material.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

***4** A hospital has an emergency backup generator which is situated 500 m away from it. The generator generates 50 kW of power at 250 V a.c. The electricity is transmitted via transmission cables with a total resistance of 0.15 Ω (Fig b).



Fig b

(a) How much power is lost in the transmission cables?

(b) (i) Find the voltage drop in the cables.

(ii) Determine whether 220-V a.c. equipment in the hospital can operate properly with the backup generator.

***5** Electricity is transmitted from the power station to the East Rail Line of MTR at 132 kV a.c.

(a) Substations are built along the railway line to step down the voltage to 25 kV a.c. (Fig c), which is then passed through the overhead transmission cables to drive the motors on the train. What is the turns ratio of the transformer in such a substation?



Fig c

(b) Electricity is transmitted to one substation via transmission cables with a total resistance of 8 Ω. The peak current value of the 132 kV a.c. is 8.5 A.

(i) What is the voltage drop in the cables?

(ii) What is the power loss in the cables for every 1000 W of power transmitted?

Ext

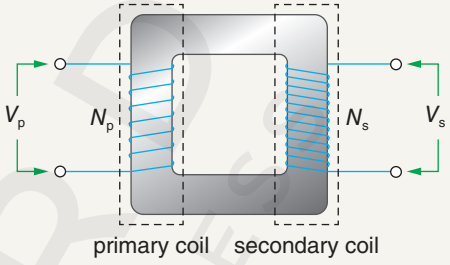
Key concepts 8

1 Transformers

- An a.c. in the primary coil produces a changing magnetic field, and hence induces an a.c. in the secondary coil.
- Voltage ratio: $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
- For a practical transformer,

  $V_p I_p \times \text{efficiency} = V_s I_s$
- For ideal transformer (efficiency = 100%),

  $\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$



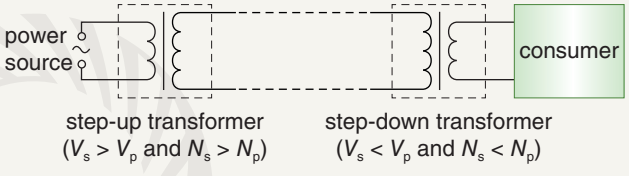
Q25 Exam Report: Poor. In (a), most candidates failed to point out that water can provide a conducting path or lower the resistance for an electric current. In (b), only a few candidates correctly explained why the human body would not get an electric shock if one of the conducting wires in the shaver circuit was touched. Most of them wrongly reasoned that the conducting wires were isolated from the high voltage. Some candidates did not understand the action of transformers and wrongly thought that current went to the earth in the primary coil because of low resistance.

2 High voltage transmission

- transmitting voltage ↑

  ⇒ current I through transmission cables ↓

  ⇒ power loss I^2R in transmission cables ↓
- is done by stepping-up the voltage before transmission and stepping-down the voltage after transmission
- a.c. is used because it can be stepped up or down easily by transformers



Key terms

1	efficiency 效率	p.328	6	step-down transformer 降壓器	p.324
2	grid system 電網系統	p.338	7	step-up transformer 升壓器	p.324
3	laminated 疊片的	p.328	8	transformer 變壓器	p.322
4	primary coil 原線圈	p.322	9	turns ratio 匝數比	p.324
5	secondary coil 副線圈	p.322	10	voltage ratio 電壓比	p.324

Ext

Revision 8

Multiple-choice questions

★1 8.1 Why is a closed iron core used in a transformer?

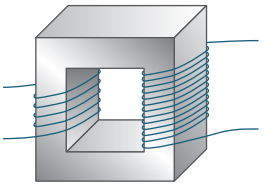


Fig a

(1) It reduces eddy currents.

(2) It reduces magnetic flux leakage.

(3) It allows current to flow from the primary coil to the secondary coil through it.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★2 8.1 A 12-V a.c. supply is connected to an ideal transformer with turns ratio 1 : 1. The secondary coil is tapped at equal intervals (Fig b).

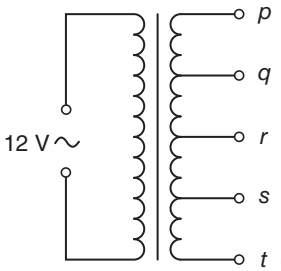


Fig b

What is the current through a '6 V, 3 W' light bulb connected across q and r?

A 0.25 A B 0.5 A

C 1 A D 2 A

★3 8.2 Which of the following statements about the transmission of electricity is/are correct?

(1) The longer the distance of transmission, the higher the power loss.

(2) The longer the length of the transmission cables, the greater the resistance of the cable per km.

(3) The higher the voltage of transmission, the greater the voltage drop in the cables.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★4 8.1 Two identical bulbs X and Y are connected to an ideal transformer as shown (Fig c).

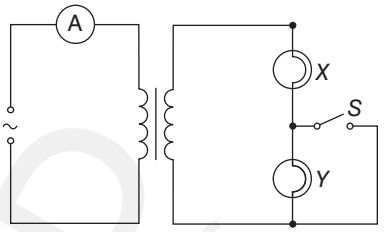


Fig c

When switch S is open, the power dissipated by X is P and the ammeter reading is I. What will they be if switch S is closed? Assume the bulbs do not burn out.

	Power dissipated by X	Ammeter reading
A	P	$0.5I$
B	$4P$	I
C	P	$2I$
D	$4P$	$2I$

(For Q5–6.) A multiple output transformer has several taps (Fig d). The turns ratio of the primary and secondary coil is 2 : 1, and the ratio of the turns in sections EF, FG and GH of the secondary coil is 1 : 3 : 16. Suppose the transformer is 95% efficient.

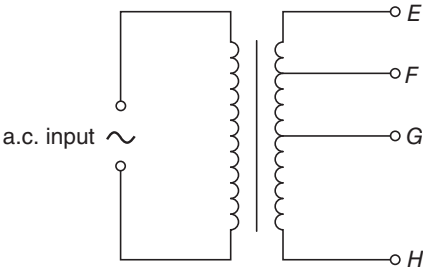


Fig d

★5 8.1 Bulb X, rated '6 V, 3 W', is connected across EF and glows normally. What is the input voltage?

A 60 V B 63 V

C 240 V D 253 V

★★6 8.1 A current of I passes through the primary circuit when bulb Y is connected across EG. What is the current thought the primary circuit if bulb Y is connected across GH? Assume bulb Y does not burn out.

A I

B $4I$

C $16I$

D $64I$

▶ Refer p.326, 333

Ext

**7 8.2 Figure e below shows a simplified electrical power transmission system. Suppose transformers T_1 and T_2 are ideal, and the total resistance of the transmission cables is R.

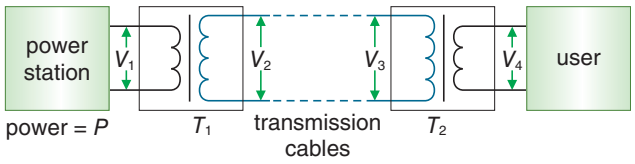


Fig e

Which of the following give(s) the power loss in the transmission cables?

(1) $\frac{P^2R}{V_2^2}$

(2) $\frac{P^2R}{(V_2 - V_3)^2}$

(3) $\frac{(V_2 - V_3)^2}{R}$

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

Refer p.339

**8 8.1

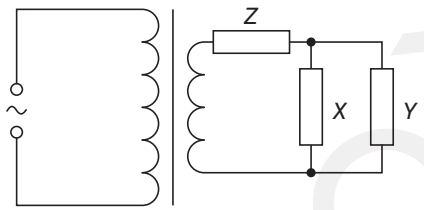


Fig f

The transformer shown in Figure f is ideal. If resistor Y is removed, which of the following will happen?

(1) The primary current decreases.

(2) The secondary current decreases.

(3) The power dissipated by resistor X increases.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

Refer p.329

9 HKDSE 2017 Paper 1A Q30 8.1

The input terminal of a transformer is connected to the 220 V mains supply. Ten identical light bulbs are connected in parallel to the output terminal of the transformer. All the light bulbs are working at their rated values of '3 V, 1.5 W'. If the efficiency of the transformer is 70%, what is the current drawn from the mains supply?

A 0.007 A (8%)

B 0.048 A (19%)

C 0.068 A (19%)

D 0.097 A (54%)

10 HKDSE 2018 Paper 1A Q30 Misc

In the circuits below (Fig g), if a 12 V sinusoidal a.c. is applied across *ab* and across *xy* respectively, the voltages across *cd* and *zw* are both 6 V. Now if a 6 V sinusoidal a.c. is applied across *cd* and across *zw* respectively, what would be the voltages across *ab* and *xy* respectively?

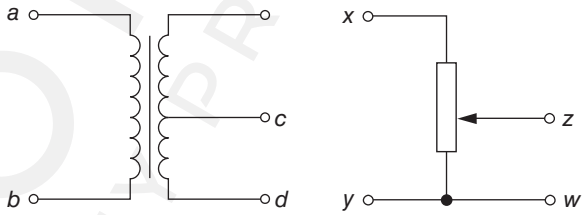


Fig g

	voltage across *ab*	voltage across *xy*
A	12 V	12 V (19%)
B	12 V	6 V (46%)
C	6 V	6 V (14%)
D	12 V	0 V (21%)

11 HKDSE 2020 Paper 1A Q29 8.1

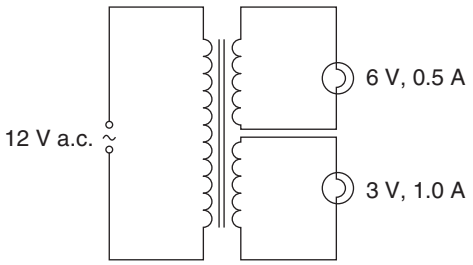


Fig h

The figure shows an ideal transformer with two secondary coils connected to two light bulbs marked '6 V, 0.5 A' and '3 V, 1.0 A' respectively. When a 12 V a.c. supply is connected to the primary coil, the bulbs work at their respective rated values. Estimate the current in the primary coil.

A 0.25 A

B 0.50 A (56%)

C 0.75 A

D 1.0 A

Ext

Conventional questions

*12 8.2 Three transmission systems, the Yunnan-Guangdong HVDC, CLP and HK Electric, are shown in Table a.

Transmission system	Transmission voltage / kV
Yunnan-Guangdong HVDC	800
CLP	400
HK Electric	275

Table a

(a) Assume the same power of 180 MW is transmitted in each system. Calculate the current through the transmission cables in each system. (4 marks)

(b) Assume the resistance of the transmission cable is 5 Ω per 100 km. Calculate the percentage of power loss per 100 km in each system. (4 marks)

(c) Explain why the Yunnan-Guangdong HVDC system transmits power at a higher voltage than the HK Electric system. (2 marks)

*13 8.1 The rated voltage and current of a printer are 110 V and 0.6 A respectively (Fig i). To operate at its rated voltage in Hong Kong, the printer is connected to the 220-V mains via a transformer with efficiency 98%.

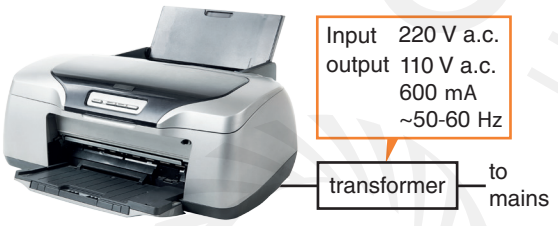


Fig i

(a) What is the turns ratio of the transformer? (2 marks)

(b) Find the peak current flowing in the printer. (2 marks)

(c) Find the power drawn from the mains when the printer is operating. (2 marks)

(d) Why must the printer be used with the transformer when connected to the mains in Hong Kong? (2 marks)

*14 8.1 Auto ignition gas stove (Fig j) requires electricity to work. The built-in gas igniter is basically a simple step-up transformer. A sudden change in the primary current can result in a high voltage (up to 15 000 V) across the secondary coil and this voltage can produce sparks to ignite the gas (Fig k).



Fig j

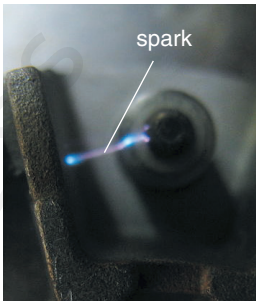


Fig k

(a) Why is a sudden change in the primary current needed, but not a slow change? (2 marks)

(b) State one possible design of the transformer that helps produce such a high voltage across the secondary coil. (1 mark)

(c) Efficient transformers usually contain laminated soft-iron cores.

(i) Why is soft iron used as the core? (2 marks)

(ii) Why is the core laminated? (2 marks)

*15 8.2 Electrical power is generated at a power station. The a.c. output from the station is first stepped up to a high voltage and then transmitted to local substations, where it is stepped down and distributed to consumers.

(a) Explain why stepping up the voltage would decrease the energy lost in transmission. (2 marks)

(b) What are the advantages of using a.c. over steady d.c. in power transmission? Explain your answer. (2 marks)

(c) State two factors that affects the efficiency of a transformer and the corresponding ways to improve the efficiency. (4 marks)

(d) Other than using high voltage to transmit electrical power, suggest another method to reduce the power loss in the transmission cables. (1 mark)

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Ext

★16 8.2 The figure below shows a model of a power transmission system (Fig l). A bulb rated at '12 V, 24 W' is connected to the output terminals. The total resistance of the cables *AB* and *CD* is 10 Ω. Assume that the two transformers are 100% efficient.

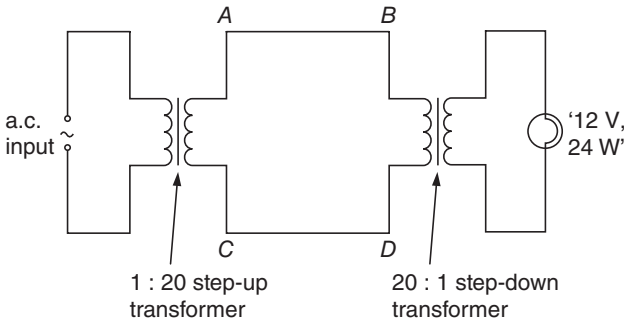


Fig l

The bulb works at its rated value.

(a) Find the current passing through the bulb. (2 marks)

(b) Find the power loss in the cables. (3 marks)

(c) Find the current drawn from the input. (1 mark)

(d) Find the voltage of the input a.c. (3 marks)

★17 8.2 Electrical power is generated at power station *A* and transmitted to the residential area *B* via the following transmission system (Fig m).

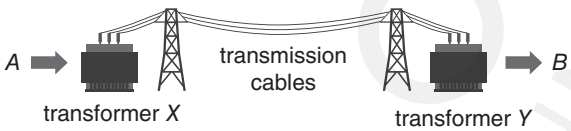


Fig m

(a) Should transformer *X* be a step-up or step-down transformer? Explain in terms of the benefit in transmission. (3 marks)

(b) Transformer *Y* steps down the voltage from 22 kV to 220 V. The total current drawn by the residential area is 50 000 A. The total resistance of the transmission cables is 25 Ω. The efficiency of transformer *Y* is 90%. Find the power loss in

(i) transformer *Y*, (2 marks)

(ii) the transmission cables. (2 marks)

(c) What is the power output of the power station? The efficiency of transformer *X* is 90%. (1 mark)

★18 Misc (a) A light bulb rated at '12 V, 50 W' is connected in series with a resistor to the 220-V mains. The bulb works at its rated values.

(i) Find the resistance of the resistor. (3 marks)

(ii) Find the power delivered by the mains. (1 mark)

(b) An identical light bulb is connected to the 220-V mains via a transformer and works at its rated value.

(i) The primary coil of the transformer has 660 turns, what is the number of turns in the secondary coil? (2 marks)

(ii) The efficiency of the transformer is 90%. Find the current and power delivered by the mains. (2 marks)

(c) Which method above is more efficient? (1 mark)

★19 Misc The resistivity of aluminium (Fig n) is 1.6 times that of copper (Fig o) and the density of copper is 3.3 times that of aluminium.

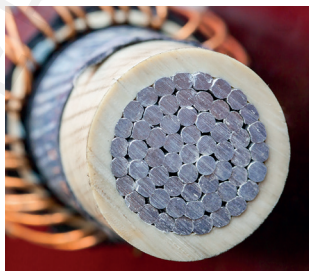


Fig n

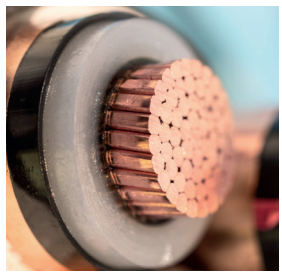


Fig o

(a) (i) A power of 10 kW is transmitted at 500 V via copper cables with a total resistance of 5 Ω. Find the power loss in the cables. (2 marks)

(ii) Find the power loss if the copper cables are replaced by aluminium cables of the same dimensions. (2 marks)

(b) (i) If two uniform wires *X* and *Y* made of aluminium and copper respectively have the same length and resistance, what is the ratio of mass of *X* to mass of *Y*? (2 marks)

(ii) Suggest a reason why aluminium is preferred over copper in making overhead transmission lines. (1 mark)

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★20 8.1 You are given a step-down transformer (Fig p), a 12 V a.c. power supply, a light bulb, two a.c. ammeters, two a.c. voltmeters and some conducting wires.

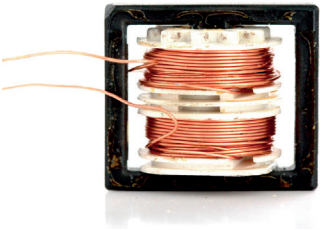


Fig p

Describe how you would use the above apparatus to estimate the efficiency of the transformer. (5 marks)

★21 8.1 You are given some copper wires with insulating coating, a soft-iron core, an a.c. power supply and two a.c. voltmeters. Describe how you would use the above apparatus to study the relationship between the voltage ratio and turns ratio of a transformer. (5 marks)

★22 8.1 Figure q shows a step-up transformer with a laminated core.

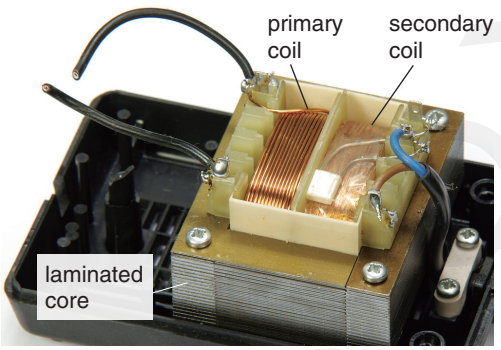


Fig q

(a) State **one** feature of the transformer that allows it to step up voltage. (1 mark)

(b) What is a laminated core? Explain how such a core can reduce the power loss in the transformer. (3 marks)

(c) Samuel suggests using a copper core is better than an iron core because copper is a better electrical conductor. Explain whether he is correct. (2 marks)

(d) The copper wire in the primary coil of the transformer is thicker than that in the secondary coil. Explain this feature. (2 marks)

★23 8.2 A power station generates electricity of power 900 kW at 11 kV, which is then stepped up to 66 kV and transmitted to a village (Fig r).

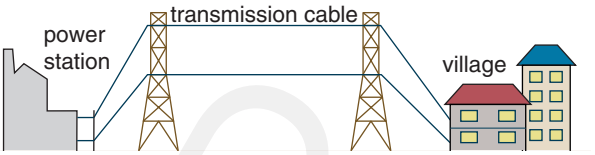


Fig r

(a) Draw a labelled diagram in Figure s to show the structure of the step-up transformer used. Find the turns ratio of the transformer. (4 marks)

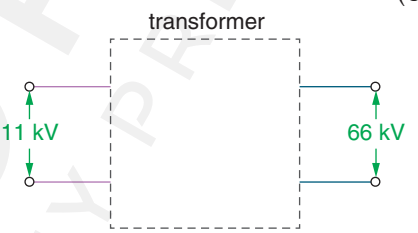


Fig s

The efficiency of the step-up transformer is 90%. The transmission cables have a total resistance of 20 Ω.

(b) Find the

(i) current,

(ii) power loss

in the transmission cables. (4 marks)

(c) What is the percentage loss in voltage across the cables? (2 marks)

(d) The voltage is stepped down to 220 V at the village. The efficiency of the step-down transformer is 90%. Find the power transmitted to the village. (2 marks)

★★24 Misc (a) State one difference between a.c. and d.c. (1 mark)

(b) Transformers are used in a.c. transmission.

(i) What is the main function of a transformer? (1 mark)

(ii) Describe how transformers are used in the transmission of electrical power from a power station in a remote district to the city. (2 marks)

(c) Figure t (p.347) shows the graph of I_p against I_s, where I_p and I_s are the primary and secondary currents of an ideal transformer.

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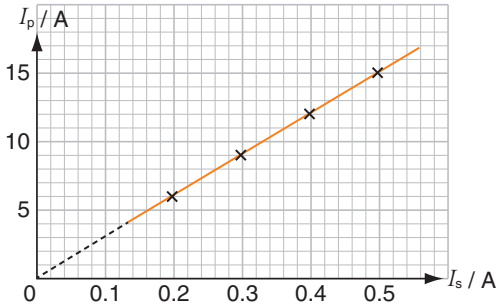


Fig t

(i) Find the turns ratio of the transformer. (2 marks)

(ii) In Figure t, sketch the graph of I_p against I_s if the transformer is not ideal. (1 mark)

(d) A number of identical lamps rated at '12 V, 24 W' are connected in parallel across the secondary coil of a transformer with a turns ratio of 20 : 1 and efficiency of 85%. The primary coil of the transformer is connected to the 240-V mains. It includes a 630-mA fuse.

(i) Calculate the maximum number of lamps that can be connected to the transformer without blowing the fuse. (2 marks)

(ii) Why is the fuse placed in the primary circuit rather than in the secondary circuit? (1 mark)

(e) Explain why a transformer does not work with a steady input voltage. (2 marks)

Refer p.329, 333

See p.341 for the Exam Report.

25 HKDSE 2012 Paper 1B Q9

8.1

Read the following description about the **shaver supply unit** in bathrooms and answer the questions that follow.

The danger of electric shock is particularly high in bathrooms. Normal electric socket outlets should not be installed in bathrooms. As electric shavers and toothbrushes are becoming popular these days, a special unit, called 'shaver supply unit' is now common in bathrooms to provide electricity just for these low power consumption electric appliances (Fig u).

The shaver supply unit consists of a transformer in which the secondary is not earthed and is completely isolated from the 220-V a.c. mains supply connecting to the primary. It can be used with 220-V or 110-V shavers.

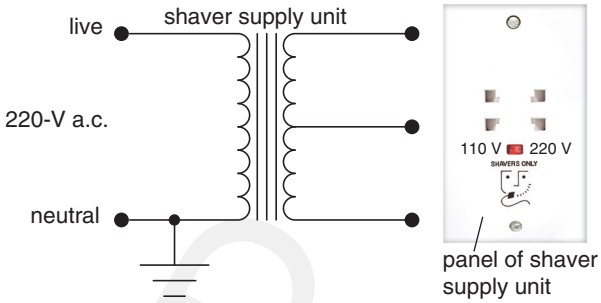


Fig u

(a) Explain why the chance of electric shock is high in bathrooms. (2 marks)

(b) Explain what would happen if the human body touches

(i) the live wire of the mains supply in the primary circuit; (2 marks)

(ii) one of the conducting wires in the shaver circuit outlet. (2 marks)

(c) What is the turns ratio of the primary coil to the secondary coil of the transformer so as to provide 110 V? (1 mark)

See p.348 for the Exam Report.

26 HKDSE 2015 Paper 1B Q8

Misc

Electricity generated from power plants are transmitted at a high voltage through overhead cables in suburban areas.

(a) Each overhead cable consists of 40 strands of identical transmission lines bundled together (Fig v).

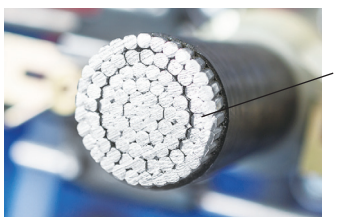


Fig v

(i) One **single** strand of transmission line has a cross-sectional area of 1.3×10^{-5} m^2 and resistivity 2.6×10^{-8} Ω m. Find the resistance **per km** of a single strand of transmission line. (2 marks)

(ii) Explain why the resistance **per km** of an overhead cable is much smaller than that of a single strand of transmission line. Estimate the resistance **per km** of an overhead cable. (2 marks)

Ext

(iii) Hence, explain why a bird can stand with both feet on a high-voltage cable without getting an electric shock (Fig w). (2 marks)



Fig w

(b) Electrical power of 180 MW is transmitted at a voltage of 4000 kV through an overhead cable.

(i) Calculate the current carried by the overhead cable. (2 marks)

(ii) Show that less than 0.1% of the electrical power is lost after transmitted through a total of 10 km of overhead cable. (2 marks)

(iii) As the voltage drip across this overhead cable is negligible, a voltage of 400 kV at the cable's end is stepped down by an ideal transformer with turns ratio 12 : 1.

(1) Find the secondary voltage from the transformer. (1 mark)

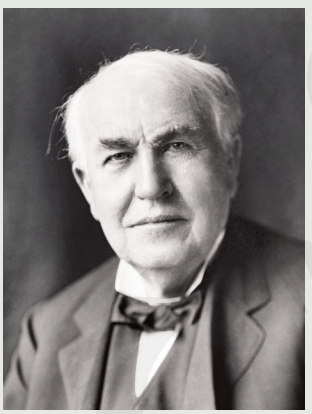
(2) State **one** factor leading to energy loss in a practical transformer and suggest the corresponding measure for improvement. (2 marks)

Know more

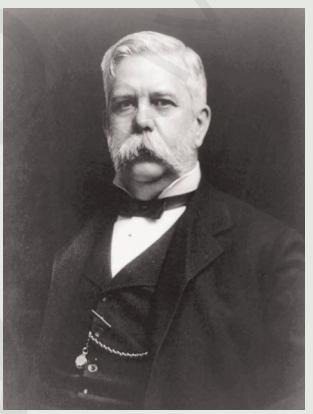
War of currents (p.337)

In 1879, Thomas Edison produced the first commercially practical incandescent light. After that, he patented a power distribution system and established power stations which provided d.c. to customers. In 1886, *George Westinghouse* built the first a.c. power system in USA. The use of a.c. spread rapidly as it could be used in long-distance transmission at a lower cost. Edison then began a campaign to discredit a.c. saying, for example, that a.c. was very dangerous to use.

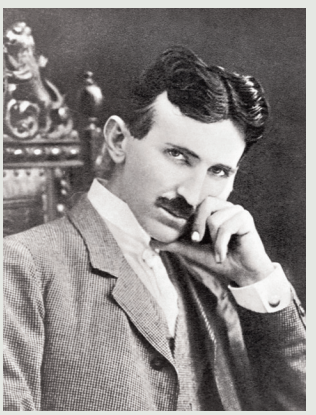
In 1888, Westinghouse acquired a key patent for an a.c. motor from Nikola Tesla and hired him as a consultant. This greatly improved the a.c. power system and the system became more popular as the industry grew, giving a.c. the final victory to this 'war of currents'.



Thomas Edison (1847–1931).



George Westinghouse (1846–1914).



Nikola Tesla (1856–1943).

OUP eResources

Q26 Exam Report: This question was well answered. In (a), a significant number of candidates misunderstood the phrase 'per km' and therefore wrongly divided the resistance by 1000 m. Some confused surface area with cross-sectional area and thus failed to find the correct answer. In (a)(iii), many candidates did not realize that electric shock is essentially due to the large amount of current passing through the body but not merely because of high voltage. Few were able to mention that negligible current would pass through the bird's body. Quite a number of them wrongly thought that the cable was covered with insulating material. Answers like 'The potential difference across the bird is zero' or 'There is no current passing through the bird's body' were common. Part (b) was well answered. A few candidates forgot to multiply the resistance per km (0.05 Ω km^{-1}) by 10 km in (b)(ii). In (b)(iii), the terms stated by some candidates were not precise, for example, 'cable' was used instead of 'wire', 'sliced iron' was used instead of 'laminated core' etc.

George Westinghouse 威斯汀豪斯